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Sneha Paul

Ph.D. Candidate

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Google Scholar

I'm a Ph.D. Candidate at Concordia University, Canada. My research focuses on multimodal unsupervised representation learning and natural language processing. A summary of my profile is as follows: published in several renowned conferences (such as CVPR, ECCV, TMLR) and journals; skilled in Python and different machine learning libraries; experienced in deep learning methods and model designing, training, and deployment.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Intern 01/2023-08/2023
BusPas Inc. Montreal, QC, CA

- Research topic: Semantic segmentation of Fish-eye images for autonomous driving.
- Developed a supervised fish-eye semantic segmentation framework based on transfer learning, published as a conference paper at ICMLA'23.
- Developed a semi-supervised fish-eye semantic segmentation framework, published and achieved the best paper award at TRB'24. The extended version of this work was published in the Journal of Imaging.

Research Intern 01/2022-03/2022
Niosense Montreal, QC, CA

- Research topic: Challenges of Data Integration in Digital Twin: A Review.
- Performed extensive literature review on the existing literature data integration for Digital Twin at road intersections.

Graduate Research Assistant 09/2021-present
Concordia Institute of Information Systems Engineering (CIISE), Concordia University Montreal, QC, CA

- Research Area: Unsupervised learning, representation learning, computer vision.

Lecturer 07/2019-06/2021
Department of Urban and Regional Planning, Khulna University of Engineering & Technology (KUET) Khulna, Bangladesh

EDUCATION

Doctor of Philosophy (Ph.D.) in Information and Systems Engineering 01/2023 — Present
Concordia University Montreal, QC, Canada

- Expected Graduation: Summer 2026
- Thesis: Unsupervised visual representation learning for special computer vision modalities.
- CGPA: 4.20/4.30 (as of Winter 2026)

Master of Applied Science (MAsc) in Quality Systems Engineering 09/2021 — 12/2022
Concordia University Montreal, QC, Canada

- Thesis: An efficient neural network architecture and training protocol for 3D point cloud classification.
- CGPA: 4.15/4.30

Bachelor of Science (BSc) in Urban and Regional Planning 04/2015 — 05/2019
Khulna University of Engineering & Technology Khulna, Bangladesh

- CGPA: 3.80/4.00
- 1st class 1st, gold medalist (selected for the upcoming graduation ceremony)

SKILLS

DL/ML Frameworks	PyTorch, TensorFlow, JAX, Keras, NumPy, SciPy, Scikit-learn
Programming Language	Python, C
Version Control & Exp. track	Git, Weights & Biases, TensorBoard
Computing, Cloud, and HPC	AWS, Google Colab, SLURM
Database and Deployment	MySQL, Oracle, Apache Spark, Docker
ML expertise	training, fine-tuning, deployment, multi-modal learning, computer vision, semi-supervised learning, self-supervised learning
Soft Skills	Technical writing (10+ publications), problem solving, critical thinking and analysis
Communication	English (IELTS band 7.5)

AWARDS AND SCHOLARSHIPS

Best Paper Award, TRB AED30 Information Systems and Technology Committee 01/2024
Transportation Research Board 103rd Annual Meeting Washington DC, USA

CIRRELT Doctoral Excellence Scholarship*Centre interuniversitaire de recherche sur les reseaux d'entreprise, la logistique et le transport (CIRRELT),***Concordia University International Tuition Award Excellence***Concordia University***Concordia University Graduate Studies Conference and Exposition Award***Concordia University***CIRRELT Masters' Excellence Scholarship***Centre interuniversitaire de recherche sur les reseaux d'entreprise, la logistique et le transport (CIRRELT),***University Gold Medal (selected)***Khulna University of Engineering & Technology,***Dean's Award for excellent performance***Khulna University of Engineering & Technology,***Academic Scholarship***Khulna University of Engineering & Technology,***01/2024***Montreal, Canada***2023***Montreal, Canada***2022, 2023***Montreal, Canada***01/2022***Montreal, Canada***Upcoming Convocation Ceremony***Khulna, Bangladesh***2016, 2018, 2019***Khulna, Bangladesh***2015-2019***Khulna, Bangladesh***PUBLICATIONS**

1. **Sneha Paul** Zachary Patterson & Nizar Bouguila, 'Domain Generalizable Adaptation of 3D Vision-Language Models via Regularized Fine-Tuning', Transaction in Machine Learning Research (**TMLR-26**).
2. **Sneha Paul** Zachary Patterson & Nizar Bouguila, 'Point Cloud as a Foreign Language for Multi-modal Large Language Model', Conference on Computer Vision and Pattern Recognition (**CVPR-26**).
3. **Sneha Paul** Zachary Patterson & Nizar Bouguila, 'An Adapter-free Fine-tuning Approach for Tuning 3D Foundation Models', International Conference on Pattern Recognition and Artificial Intelligence (**ICPRAI-26**).
4. **Sneha Paul** Zachary Patterson & Nizar Bouguila, 'Improving 3D Semi-supervised Learning by Effectively Utilizing All Unlabelled Data', European Conference on Computer Vision (**ECCV-24**). [\[Project Page\]](#)
5. **Sneha Paul**, Zachary Patterson & Nizar Bouguila, 'FishSegSSL: A Semi-supervised Semantic Segmentation Framework for Fish-eye Images', Journal of Imaging (2024).
6. **Sneha Paul**, Zachary Patterson & Nizar Bouguila, 'Semi-supervised Semantic Segmentation on Vehicle-mounted Fish-eye Camera Images', Transportation Research Board 103rd Annual Meeting (**TRB-24**).
7. **Sneha Paul**, Zachary Patterson & Nizar Bouguila, 'Semantic Segmentation Using Transfer Learning on Fisheye Images', 22nd International Conference on Machine Learning and Applications (**ICMLA-23**).
8. **Sneha Paul**, Zachary Patterson & Nizar Bouguila, 'CrossMoCo: Multi-modal Momentum Contrastive Learning for Point Cloud', 20th Conference on Robots and Vision (**CRV-23**).
9. **Sneha Paul**, Zachary Patterson & Nizar Bouguila, 'DualMLP: A Two-stream Fusion Model for 3D Point Cloud Classification', The Visual Computer Journal (2023).
10. **Sneha Paul**, Zachary Patterson & Nizar Bouguila, 'Improved Training for 3D Point Cloud Classification', IAPR Joint International Workshops on Statistical Techniques in Pattern Recognition and Structural and Syntactic Pattern Recognition (**S+SSPR-22**).
11. **Sneha Paul** & Shuvendu Roy, 'Forecasting the Average Temperature Rise in Bangladesh: A Time Series Analysis', Journal of Engineering Science.
12. Shuvendu Roy & **Sneha Paul**, 'Land-Use Detection Using Residual Convolutional Neural Network', International Conference on Advances in Science, Engineering and Robotics Technology (**ICASERT-19**).

ACADEMIC SERVICES**Exam Invigilator***Concordia University, Montreal, Canada***Reviewer**

- British Machine Vision Conference (BMVC-25)
- Computer Vision and Pattern Recognition (CVPR-25)
- International Conference on Learning Representations (ICLR-25)
- European Conference on Computer Vision (ECCV-24)
- International Conference on Machine Learning and Applications (ICMLA-23)
- International Conference on Robots and Vision (CRV-23)
- Pattern Recognition
- IEEE Transactions on Industrial Informatics (TII)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- Engineering Applications of Artificial Intelligence